

Introduction and Classical Cryptography

CS 171: Cryptography

Sanjam Garg

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Overview

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- **Course website:** <https://sanjamg.github.io/cs171-fall126/> (schedule, readings, policies, and links to everything below).
- **Lectures:** Tue/Thu, 12:30–2:00pm, Moffitt Library 145.
- **Ed:** all course announcements and Q&A — please join!
- **Gradescope:** homework and exams are submitted here (Course ID 1362375, entry code VEG54G).

- **Instructor:** Sanjam Garg (sanjamg@berkeley.edu).
- **Teaching Assistants:**
 - Diogo Antunes (dsantunes@berkeley.edu)
 - Zoë Ruha Bell (zbell@berkeley.edu)
- **DSP accommodations:** please have your letter sent to the instructor within the first two weeks, and contact the course staff about homework/exam logistics.

Prerequisites

- The only formal prerequisite is discrete mathematics (**CS 70**).
- We assume familiarity with basic (discrete) **probability** and **modular arithmetic**.
- Some exposure to algorithms: comfortable reading pseudocode and familiar with big- O notation.

Homework

- Homeworks (and quizzes) are submitted on Gradescope, due at **8:59pm** on the due date.
- **Late policy:** lose 5% for every hour late — no exceptions.
(1 minute late $\Rightarrow$ -5%; 61 minutes late $\Rightarrow$ -10%.)
- **Drop policy:** best $x - 3$ out of x homeworks (e.g. best 7 out of 10).
- **Collaboration:** discussing problems with classmates is encouraged, but you must write up solutions *on your own*, never possess anyone else's solutions, and acknowledge all collaborators and sources.

Full policies (late submissions, collaboration, honor code) are on the course website.

Exam Dates and Policies

- **Midterm I:** Thu Sep 24 (during class, Moffitt Library 145).
- **Midterm II:** Thu Oct 29 (during class, Moffitt Library 145).
- **Final:** Fri Dec 18, 8am–11am (location TBD).
- **Oral exams:** one week before each midterm, and during the last week of classes (before the final).
- All exams are **mandatory** — please keep these time slots free.

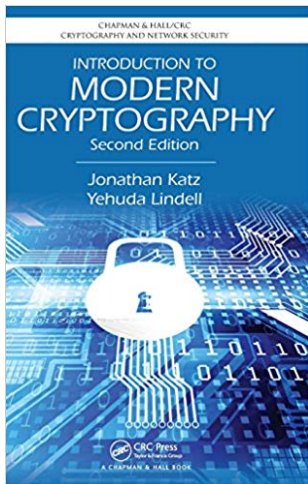
Grading

- Homeworks and quizzes — 10%
- Midterm I — 25%
- Midterm II — 25%
- Final — 40%

Clobber policy (lowest midterm)

Your lowest midterm score is replaced by the max of that score and your final-exam z-score mapped onto that midterm. Your other midterm is unchanged.

Homework scores could potentially be normalized based on the oral exams.



- Katz and Lindell, *Introduction to Modern Cryptography* ("KL"). Any edition is fine.
- Readings for each lecture are listed on the course website.
Today: KL 1.1–1.3.
- Other excellent (mostly free) resources are linked from the website, e.g. Boneh–Shoup, *A Graduate Course in Applied Cryptography*, and Rosulek, *The Joy of Cryptography*.

Course Roadmap

- **Weeks 1–5: Private-key encryption.** Perfect secrecy and the one-time pad, pseudorandomness, PRFs, CPA/CCA security, stream and block ciphers (SPNs, Feistel, DES).

(Midterm I: Sep 24)

- **Weeks 6–7: Integrity and hashing.** MACs, authenticated encryption, hash functions, one-way functions.
- **Weeks 8–10: Public-key cryptography.** Groups and hardness assumptions, Diffie–Hellman, El Gamal, RSA, digital signatures and PKI.

(Midterm II: Oct 29)

- **Weeks 11–15: Advanced topics.** Pairings and IBE, commitments, zero-knowledge proofs, succinct proofs, secret sharing, multiparty computation.
- **Final:** Dec 18.

Questions?

New material!

Please ask questions throughout this class.

About the Course

- Cryptography is used **everywhere**, with many applications.
- **Course goal**: learn the theoretical basis of the cryptography used in the real world.
 - Learn about key **primitives**.
 - Understand what **security** they provide.
 - Know how to **use** them.
 - Understand “how things work”.
- **Do not**: brew your own crypto!

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Crypto mindset

Think carefully about definitions, assumptions, and proofs — not ad hoc tricks.

Cryptography Historically

- Focused on **private communication** between two parties using previously shared secret information.
- Lack of clarity on what it means to be secure.
- Heuristic, unprincipled design approach.
- Break-and-fix cycle (cat-and-mouse game).



Modern Cryptography

- More than just confidentiality, e.g. data integrity.
- More tasks: **public-key** cryptography.
- **Rigorous** security definitions.
- Sound mathematical principles for arguing/proving security.
 - E.g. basing it on **factoring**.



From Art to Science

In summary

Cryptography has developed from an **art** to a **science**.

Classical Cryptography

- Focused on secret-key / shared-key / private-key / symmetric-key cryptography.
- Start with simple schemes used for centuries.
- Demonstrate the dangers of an unprincipled approach.
- Show why simple tricks are unlikely to work.

Private-Key Encryption (Confidential Communication)



Private-Key Encryption (Confidential Communication)

key
↓
 k

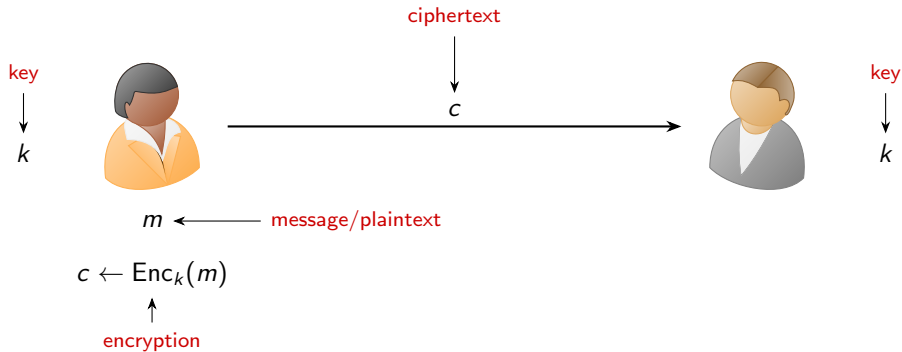


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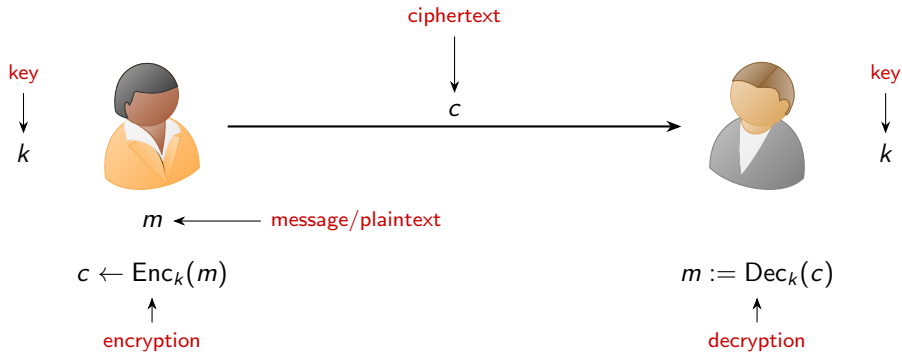
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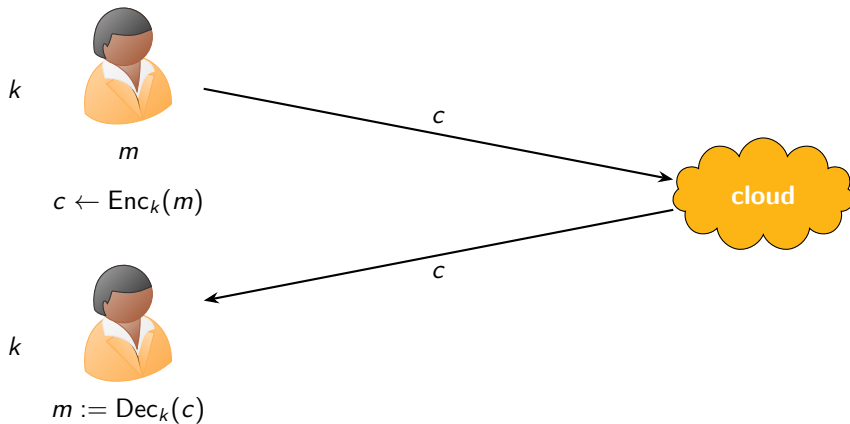
Private-Key Encryption (Confidential Communication)



Private-Key Encryption (Confidential Communication)



Private-Key Encryption (Confidential Self-Communication)



Private-Key Encryption (Syntax)

Definition 1 (Private-Key Encryption Scheme)

A **private-key encryption scheme** is a message space $\mathcal{M}$, a key space $\mathcal{K}$, and algorithms (Gen, Enc, Dec):

- Gen (key generation): outputs $k \in \mathcal{K}$.
- Enc (encryption): on key k and message $m \in \mathcal{M}$, outputs ciphertext $c \leftarrow \text{Enc}_k(m)$.
- Dec (decryption): on key k and ciphertext c , outputs $m := \text{Dec}_k(c)$ (or “error”).

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Correctness

For all $m \in \mathcal{M}$ and all k output by Gen: $\text{Dec}_k(\text{Enc}_k(m)) = m$.

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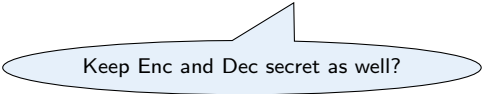
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Keep Enc and Dec secret as well?

Kerckhoffs's Principle

Kerckhoffs's principle

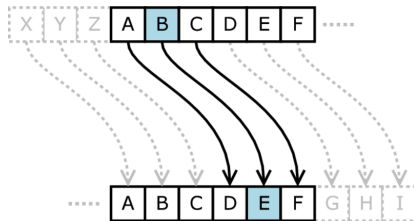
The cipher method must not be required to be secret, and it must be able to fall into the hands of the enemy without inconvenience.

- Only the **key** k is kept secret.
- Very important in modern cryptography: use open, well-studied tools.



The Shift Cipher

- Consider encrypting English text.
- Associate 'a' with 0, 'b' with 1, ..., 'z' with 25.
- Key space $\mathcal{K} = \{0, \dots, 25\}$.
- To encrypt with key k , shift every letter by k positions (with wraparound, mod 26).
- Decryption reverses this.



Example shift by $k = 3$.

The Shift Cipher, Formally

Definition 2 (Shift Cipher)

- $\mathcal{M} = \{\text{strings over the lowercase English alphabet}\}.$
- $\mathcal{K} = \{0, \dots, 25\}.$
- Gen: choose uniform $k \in \{0, \dots, 25\}.$
- $\text{Enc}_k(m_1 \cdots m_t)$: output $c_1 \cdots c_t$, where $c_i := [m_i + k \bmod 26].$
- $\text{Dec}_k(c_1 \cdots c_t)$: output $m_1 \cdots m_t$, where $m_i := [c_i - k \bmod 26].$

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- $\text{Dec}_k(c_1 \cdots c_t)$: output $m_1 \cdots m_t$, where $m_i := [c_i - k \bmod 26].$

Correctness

For each i :

$$[c_i - k \bmod 26] = [[m_i + k \bmod 26] - k \bmod 26] = [m_i + k - k \bmod 26] = m_i.$$

Is the Shift Cipher Secure?

Ciphertext

etarvqogcpuetarvqitcrja

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Ciphertext

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- **No** — only 26 possible keys!
 - Try decrypting with every possible key.
 - Only one possibility will “make sense”.
 - Assumes English text has enough structure/redundancy that only one candidate makes sense.
- Example of a **brute-force** (exhaustive-search) attack.

Summary

- Cryptography has moved from an **art** to a **science**: definitions, assumptions, proofs.
- **Private-key encryption** syntax: (Gen, Enc, Dec) with correctness.
- **Kerckhoffs's principle**: security must rest only on the secret key.
- The **shift cipher**: only 26 keys, so a **brute-force** attack tries them all.

Reading

Today: KL 1.1–1.3. Next time: breaking more classical ciphers and their lessons, principles of modern cryptography, and perfect secrecy (KL 1.3–1.4, 2.1).